Notes for the Donaldson-Kronheimer learning seminar at UT Austin, Feb 2026.

1 Setup

Let E be a bundle over a compact, oriented Riemannian four-manifold X. The bundle has a space of connections $\mathscr{A}$, an affine space over the space of $\mathfrak{g}$-valued $1-$forms $\Omega^1(\mathfrak{g}_E)$.

The gauge group (i.e. space of bundle automorphisms) $\mathscr{G}$ acts on $\mathscr{A}$ by the local formula

$$u^*A = u^{-1}du + u^{-1}Au.$$

Our first goal is to find a local model for the orbit space $\mathscr{A}/\mathscr{G}$. That is, for a given connection A, what does quotienting by $\mathscr{G}$ do to a local neighborhood of A? We will see that this orbit space naturally has the structure of a *stratified* manifold.

Afterwards, we will analyze the subspace of this orbit space corresponding to ASD connections. By writing this subspace as the kernel of a Fredholm operator, we will see that the Yang-Mills moduli space is a finite-dimensional manifold with dimension given by the Atiyah-Singer index theorem.

2 The orbit space

We first forget the specific setup of our problem and seek to generally understand what quotienting a manifold by a continuous group action looks like locally around a point.

Toy Example 2.1: $\mathbb{R} \times S^1 \curvearrowright \mathbb{R}^3$

Consider the action of $G = \mathbb{R} \times S^1$ on $V = \mathbb{R}^3$, where the factor of $\mathbb{R}$ translates V along the z-axis while the factor of S^1 rotates V about the z-axis.

At the origin, the derivative of the action $G \times V \to V$ with respect to the G-factor (i.e. a map $d_G : T_0G \to T_0V = V$ has image exactly the z-axis. Hence acting by G identifies vertical fibres of V, and we can identify the remaining space with

$$(\text{Im } d_G)^{\perp} = \text{the } \langle x, y\rangle\text{-plane}.$$

However, we are not done: we still have the quotient out the $\langle x, y\rangle$-plane by the *stabilizer* of 0, which in this case is S^1. Hence our local model for a neighborhood of 0 under the group action is

$$U_{[0]} \cong \mathbb{R}^2/\{\text{rotation about } 0\} \cong [0, \infty).$$

Defining $U_{[0]}$ as this quotient naturally gives it the structure of a stratified space, with different strata corresponding to the size of the stabilizer at each point. Here the strata would be

$$F_0 = \{0\}$$
$$F_1 = (0, \infty).$$

2.1 What does this framework look like when applied to $\mathscr{A}/\mathscr{G}$?

Analytical setup. In order to be able to easily talk about small local neighborhoods, we will give $\mathscr{A}, \mathscr{G}$ Hilbert space structure by letting $\mathscr{A}$ be the space of L^2_{l-1} connections and $\mathscr{G}$ the space of L^2_l gauge transformations (Note that there is a du term when $u \in G$ acts on $A \in \mathscr{A}$ so $\mathscr{G}$ has to have one more level of regularity). Each choice of regularity l gives us a quotient space $\mathscr{A}/\mathscr{G}$ modeled on a different Banach space. Luckily this regularity ends up not mattering once we restrict to the space of ASD connections.

Now $\mathscr{A}, \mathscr{G}$ are smooth Banach manifolds (around every point there is a neighborhood homeomorphic to a Banach space and transition functions are smooth) and the action $\mathscr{G} \times \mathscr{A} \to \mathscr{A}$ is a smooth map of Banach manifolds. At a point $A \in \mathscr{A}$, we can compute the derivative of the action in the $\mathscr{G}$ variable: For a small perturbation $I + \eta$ of the identity,

$$\begin{aligned}(I+\eta)^* A &= (I-\eta)d(I+\eta) + (I-\eta)A(I+\eta) \\ &= A + (d\eta + A\eta - \eta A) + O(\eta^2).\end{aligned}$$

Pulling back η to a Lie algebra-valued function $e \in \Omega^0(\mathfrak{g}_E)$ along the exponential map, we see that the derivative of the action $D_{\mathscr{G}}|_A : \Omega^0(\mathfrak{g}_E) \to \Omega^1(\mathfrak{g}_E)$ is exactly the covariant derivative with respect to A,

$$D_{\mathscr{G}}|_A(e) = de + [A, e] = d_A(e).$$

> **Note** (I'm not sure this is right?).
>
> $A \in G$ acts on the Lie algebra of G by the adjoint action $e \mapsto Ae - eA$ and so the covariant derivative $d_A = d + A$ acts on elements $e \in \mathfrak{g}$ of the Lie algebra as $d_A e = de + Ae - eA$.

Hence in a local neighborhood of A, the action of $\mathscr{G}$ identifies points in the image of d_A. Elliptic theory gives us the decomposition

$$\Omega^1(\mathfrak{g}_E) = \text{Im } d_A + \ker d_A^*$$

and so we can write a neighborhood of the orthogonal complement as

$$T_{A,\epsilon} = \{a \in \Omega^1(\mathfrak{g}_E) \mid d_A^* a = 0, ||a||_{L^2_{l-1}} < \varepsilon\}.$$

For a point $A \in \mathscr{A}$ we'll write the stabilizer of A as

$$\Gamma_A = \{u \in \mathscr{G} \mid u(A) = A\}.$$

Just as in the toy example, we can now write down a local model for the quotient space $\mathscr{A}/\mathscr{G}$.

Proposition 2.3: Local structure model

For small ε the quotient map $\mathscr{A} \to \mathscr{A}/\mathscr{G}$ induces a homeomorphism from the quotient $T_{A,\varepsilon}/\Gamma_A$ to a neighborhood of $[A]$ in $\mathscr{A}/\mathscr{G}$.

2.2 The stabilizers Γ_A

Note (3 ways to view a connection).

Recall from John's talk that there are three equivalent ways we can view a connection $A \in \mathscr{A}$:

1. As a covariant derivative d_A
2. As a matrix-valued 1-form $A \in \Omega^1(\mathfrak{g}_E)$ given by $d_A = d + A$
3. As specifying an identification of fibers E_x, E_y for $x, y \in X$ by horizontally lifting of over a path $\gamma : [0,1] \to X$ from x to y.

We will use the three interchangeably in this section.

Fix a connection $A \in \mathscr{A}$ and consider its stabilizer $\Gamma_A \subset \mathscr{G}$. Notice that if $u \in \mathscr{G}$ preserves A then it must preserve parallel transport under A. Hence it is enough to specify how u acts on a single fiber E_x: the action on all other fibers is determined by parallel transport starting from E_x. In fact we have a further restriction: u must preserve parallel transport from E_x to *itself* along nontrivial closed loops. That is, the action of u on E_x must commute with elements in the holonomy group $H_A \subset G$.

Note (Definition of the holonomy group H_A).

Any two points x, y in the same connected component of X have holonomy groups $H_A(x), H_A(y)$ which are conjugate in G. Hence the holonomy group H_A is a well-defined subgroup of G up to conjugacy.

Conversely, since the data of a connection A is precisely the data of parallel transport under A, any $u \in \mathscr{G}$ preserving parallel transport must be in Γ_A. In summary:

Proposition 2.5.1. *For any connection A over a connected base X, Γ_A is isomorphic to the centralizer of H_A in G.*

2.3 Stratification of $\mathscr{A}/\mathscr{G}$

We partition $\mathscr{B} = \mathscr{A}/\mathscr{G}$ into a disjoint union of pieces $\mathscr{B}^\Gamma$ labelled by the conjugacy classes of the stabilizer groups $\Gamma_A \subset G$. This partition in fact makes $\mathscr{B}$ into a *stratified* space.

Definition 2.5.1 (Informal)**.** *A* ***stratification*** *of a topological space V is a decomposition of V into strata*

$$V = \bigsqcup_{X \in S} X$$

where:

1. *Each X is a smooth, locally closed (i.e. X open in $\overline{X}$) submanifold of V*
2. *If $Y \cap \overline{X} \neq \emptyset$, then $Y \subseteq \overline{X}$.*

In particular consider the top stratum. The stabilizer Γ_A of any connection A always contains at least the center $Z(G)$ of G. Let

$$\mathscr{A}^* = \{A \in \mathscr{A} \mid \Gamma_A = Z(G)\}$$

be the open subset of $\mathscr{A}$ consisting of connections whose stabilizer is minimal.

If $u \in \Gamma_A$ is in fact in $Z(G)$, then we can check that u preserves *all* connections.

Note.

Proof.

$$\begin{aligned} u \in \Gamma_A &\implies 0 = d_A u = u^{-1} du + \cancel{Au - uA} \\ &\implies u(B) = u^{-1} du + u^{-1} B u = B. \end{aligned}$$

□

Thus for $A \in \mathscr{A}^*$, Γ_A acts trivially on all of $\mathscr{A}$. By 2.3 this means $\mathscr{B}^*$ is modelled locally on $T_{A,\varepsilon}$, which are ε-balls in the Hilbert space $\ker d_A^* \subset L^2_{l-1}(\Omega^1(\mathfrak{g}_E))$.

Computation 2.7: $SU(2)$ strata

Suppose $G = SU(2)$ so that E is an $SU(2)$-bundle over X. The quotient space $\mathscr{B} = \mathscr{A}/\mathscr{G}$ is partitioned into stratum $\mathscr{B}^\Gamma$ where each Γ is the centralizer of some holonomy group $H_A \subset SU(2)$. Although $SU(2)$ has many possible subgroups, up to conjugacy the only possible centralizers of those subgroups are:

- $\pm I = Z(SU(2))$
- S^1
- $SU(2)$.

The first case $\mathscr{B}^*$ corresponds to the generic stratum.

In the second case, consider a single fiber E_x. we can always choose a basis for $SU(2)$ as a matrix group so that the S^1 subgroup looks like

$$S^1 = \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \mid \theta \in \mathbb{R} \right\}.$$

The connections stabilized by this subgroup are exactly the diagonal connections

$$A = \begin{pmatrix} ia & 0 \\ 0 & -ia \end{pmatrix}, \; a \in \Omega^2(X, \mathbb{R}).$$

Hence the fiber E_x splits in a way which is compatible with the connection. By parallel transport, this picture extends globally (?), giving us a splitting of the vector bundle

$$E = L \oplus L^{-1}.$$

Conversely, any connection compatible with such splitting a is stabilized by an S^1-subgroup given by the above local picture. Hence **the stratum $\mathscr{B}^{S^1}$ corresponds to connections compatible with a splitting of the vector bundle**.

The third case corresponds to the trivial product connection. (We will work with nontrivial bundles so won't have to worry!)

3 The ASD Moduli Space

We now look at the ASD Moduli space: solutions to the ASD equation $F^+(A) = 0$ modulo gauge equivalence.

Analytical setup. Let $\mathscr{M}(l) \subset \mathscr{B}(l)$ denote the space of L^2_{l-1} ASD connections modulo L^2_l gauge transformations.

By an elliptic regularity argument, we can show that if A is an ASD connection of class L^2_{l-1}, there is a L^2_l gauge transformation u such that $u(A)$ is in L^2_l. This shows that the natural inclusion $\mathscr{M}(l+1) \hookrightarrow \mathscr{M}(l)$ is surjective, and in fact it turns out to be a homeomorphism. Hence we can unambiguously refer to the moduli space $\mathscr{M} = \mathscr{M}_E$ and drop the Sobolev index.

Main Idea. Locally around an ASD connection A, the positive part of the curvature operator F^+ is a Fredholm map, allowing us to compute the dimension of its kernel.

Let A be an ASD connection and define

$$\psi : T_{A,\varepsilon} \to \Omega^+(\mathfrak{g}_E), \quad \psi(a) = F^+(A + a) = d_A^+ a + (a \wedge a)^+.$$

Let $Z(\psi) \subset T_{A,\varepsilon}$ be the zero set of ψ. In M, a neighborhood of $[A]$ locally looks like $Z(\psi)/\Gamma_A$.

Note (Definition of Fredholm maps).

Definition 3.1.1. *A bounded linear map $L : U \to V$ between Banach spaces is* Fredholm *if it has finite dimensional kernel and cokernel. The* index *of a Fredholm map is the difference of the* *dimensions:*

$$\mathrm{ind}(L) = \dim\ker L - \dim \mathrm{coker}\ L.$$

The index of a Fredholm map is unchanged by continuous deformations of L through Fredholm operators. Because of this, we can extend the Fredholm definition to smooth maps and still get a well-defined index.

Definition 3.1.2. *Let N be a connected open neighborhood in the Banach space U. A smooth map $\phi : N \to V$ is called Fredholm if for each point $x \in N$ the derivative*

$$(D\phi)_x : U \to V$$

is a linear Fredholm operator. In this case the index of $(D\phi)_x$ is independent of x and is referred to as the index of ϕ.

Returning to 3, we will show that ψ is a Fredholm operator and compute its index. The derivative of ψ at 0 is

$$d_A^+ : \ker d_A^* \to \Omega^+(\mathfrak{g}_E).$$

From Jemma's talk, we know that $d_A^* \oplus d_A^+ = \delta_A : \Omega^1 \to \Omega^0 \oplus \Omega^+$ is elliptic, hence Fredholm. This descends to $\ker d_A^+|_{d_A^*}$ being Fredholm by the following computation.

Have

$$\begin{aligned} \ker \delta_A &= \ker d_A^* \cap \ker d_A^+ = \ker(d_A^+|_{\ker d_A^*}) \\ \mathrm{coker}\ \delta_A &\supseteq \mathrm{coker}\ d_A^+ \text{ [check this?]} \end{aligned}$$

so the restriction of d_A^+ to the kernel of d_A^* is Fredholm with index

$$\begin{aligned} \mathrm{ind}\ \psi &= \dim\ker d_A^+|_{d_A^*} - \dim\mathrm{coker}\ d_A^+|_{d_A^*} \\ &= \dim\ker\delta_A - (\dim\mathrm{coker}\ \delta_A - \dim\mathrm{coker}\ d_A^*) \\ &= \mathrm{ind}\ \delta_A + \dim\ker d_A \\ &= \mathrm{ind}\ \delta_A + \dim\Gamma_A \end{aligned}$$

Using that $\mathrm{coker}\ d_A^* = \ker d_A$ by definition of adjoint and that $u \in \Gamma_A \iff d_A u = 0$.

Since δ_A is elliptic, we can apply the Atiyah-Singer index theorem. For $SU(2)$-bundles E this takes the form

$$s := \mathrm{ind}\ \delta_A = 8c_2(E) - 3(1 - b_1(X) + b_2^+(X)).$$

(Here b_2^+ is the rank of the largest positive definite subspace of H^2 under the intersection form.)

For a generic connection $A \in \mathscr{A}^*$, $\dim\Gamma_A = 0$, and in fact (at least for our examples?? no idea how this works) a Weitzenbock formula shows that d_A^+ is surjective. Hence we have

$$\dim\mathscr{M} = \dim\ker d_A^+|_{\ker d_A^*} = \mathrm{ind}\ \delta_A = s.$$

3.1 Examples

Note (Structure of $SU(2)$-bundles).

$SU(2)$-bundles over compact 4-manifolds are classified by their second Chern class $c_2(E)$ (note this classification fails over higher dimensions!). By the Whitney product formula, an $SU(2)$ bundle E splits as a $E = L_1 \oplus L_2$ if and only if $-c_2(E)$ is a square in the cohomology ring:

$$-c_2(E) = \alpha \smile \alpha, \quad \alpha = c_1(L).$$

If so, the number of splittings corresponds to half the number of solutions to 3.2.

1. Let E be the $SU(2)$-bundle over S^4 with $c_2(E) = 1$. The Atiyah-Singer index formula gives $\dim \mathscr{M} = s = 5$, and since S^4 has no second cohomology E has no splittings. In fact by an explicit construction (the ADHM construction) the moduli space turns out to be the open 5-ball B^5.

2. Let E be the $SU(2)$-bundle over $\overline{\mathbb{CP}^2}$ with $c_2(E) = 1$. $\overline{\mathbb{CP}^2}$ has intersection form (-1), so $b_2^+ = 0$. The Atiyah-Singer index formula gives $\dim \mathscr{M} = s = 5$. Here $\alpha \smile \alpha = -c_2(E) = -1$ has one solution (up to sign), and so there is one singular point A. Around the singular point, $\ker d_A^* = \dim \mathscr{M} + 1 = 6$ and so a local neighborhood looks like $\mathbb{C}^3/S^1 \cong$ cone over $\overline{\mathbb{CP}}^2$.